MATERIALS INFORMATICS REPORT

Prediction and Optimization of Aerospace Aluminum Alloy Strength Using Machine Learning

Course: Materials Informatics

Topic: Aerospace Materials Challenge

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ABSTRACT

The aerospace industry continuously seeks lightweight materials with superior mechanical performance to improve fuel efficiency, structural reliability, and safety. Aluminum alloys are among the most widely used aerospace materials due to their excellent strength-to-weight ratio and corrosion resistance. However, predicting the performance of these alloys under varying compositions and processing conditions is a complex task.

This project presents a materials informatics approach for predicting the tensile strength of aerospace aluminum alloys using machine learning. A dataset consisting of alloy composition and heat-treatment parameters was created and analyzed using Linear Regression implemented in Python. Data preprocessing techniques including train-test splitting and feature selection were applied before model training.

The model demonstrated strong prediction capability with a high R^2 score and low RMSE value. Based on the prediction results, an optimized alloy composition suitable for aerospace applications was proposed. This study demonstrates how machine learning can accelerate material optimization and reduce the need for expensive experimental testing in aerospace engineering.

1. INTRODUCTION

Modern aerospace engineering depends heavily on advanced materials capable of withstanding high stress, vibration, fatigue loading, and thermal fluctuations. The selection of suitable materials directly influences aircraft performance, fuel efficiency, and safety.

Aluminum alloys remain one of the most important aerospace materials because of their lightweight nature and excellent mechanical properties. However, developing optimized alloys through traditional experimental methods requires significant time and cost.

Materials Informatics combines materials science with machine learning and data analytics to accelerate material discovery and optimization. In this project, a machine learning model is developed to predict tensile strength based on alloy composition and heat-treatment conditions.

2. PROBLEM DEFINITION

The aerospace industry requires lightweight structural materials with high tensile strength and durability. The challenge is to predict and optimize the tensile strength of aerospace aluminum alloys using machine learning techniques.

The target property selected for this project is tensile strength because it directly affects structural reliability and load-bearing capacity in aircraft applications.

3. OBJECTIVES

1. To study aerospace aluminum alloys and their applications.
2. To create a dataset containing alloy composition and processing parameters.
3. To preprocess the dataset for machine learning analysis.
4. To implement Linear Regression using Python.
5. To evaluate the model using RMSE and R^2 metrics.
6. To propose an optimized aerospace alloy composition.

4. SCIENTIFIC BACKGROUND

Aerospace aluminum alloys mainly belong to the 2xxx, 6xxx, and 7xxx series. These alloys provide high strength while maintaining low density. Alloying elements such as magnesium, copper, and zinc significantly influence mechanical behavior.

Magnesium improves hardness and strength, copper enhances precipitation hardening, and zinc contributes to superior tensile strength. Heat treatment also plays a critical role in modifying microstructure and improving mechanical properties.

5. DATASET CREATION

A dataset containing 30 data points was prepared using alloy composition and heat-treatment parameters. The dataset features included magnesium percentage, copper percentage, zinc percentage, heat-treatment temperature, and aging time. The target property was tensile strength measured in MPa.

The dataset satisfied the minimum requirement mentioned in the assignment guidelines.

6. DATA PREPROCESSING

Data preprocessing involved checking missing values, selecting relevant features, and splitting the dataset into training and testing sets. The dataset was divided into 80% training data and 20% testing data.

Feature selection was performed to ensure that only parameters affecting tensile strength were included in the machine learning model.

7. MACHINE LEARNING MODEL

Linear Regression was selected as the machine learning model because it is simple, interpretable, and effective for numerical prediction problems. The model was trained using alloy composition and processing parameters to predict tensile strength.

Python libraries including Pandas, NumPy, Scikit-learn, and Matplotlib were used for implementation in Jupyter Notebook.

8. MODEL EVALUATION

The model was evaluated using R^2 score and RMSE.

Obtained R^2 Score: 0.958

Obtained RMSE: 4.658 MPa

The high R^2 score indicates strong prediction capability, while the low RMSE value shows that prediction error is minimal.

9. RESULTS AND DISCUSSION

The machine learning model successfully predicted tensile strength based on alloy composition and heat-treatment parameters. Increasing zinc and magnesium content generally improved tensile strength. Heat-treatment temperature also showed a significant influence on precipitation hardening.
The prediction results closely followed expected aerospace material trends, demonstrating the effectiveness of machine learning in materials engineering.

10. OPTIMIZED MATERIAL DESIGN

Based on model predictions, the following optimized composition was proposed:

- Magnesium (Mg): 2.9%
- Copper (Cu): 1.9%
- Zinc (Zn): 6.2%
- Heat Treatment Temperature: 138°C
- Aging Time: 14 hours

The predicted tensile strength for the optimized alloy was approximately 580 MPa, making it suitable for lightweight aerospace structural applications.

11. ADVANTAGES OF MATERIALS INFORMATICS

Materials Informatics reduces experimental cost and accelerates material discovery. Machine learning enables engineers to identify optimized compositions quickly and improves the efficiency of aerospace material development.

This approach also reduces trial-and-error experimentation and supports intelligent material design.

12. LIMITATIONS

Although the project achieved good prediction accuracy, certain limitations exist. The dataset size was relatively small, and only a limited number of processing parameters were considered. Experimental validation was not performed.
Future work can include larger datasets and advanced machine learning techniques such as Artificial Neural Networks and Support Vector Machines.

13. FUTURE SCOPE

Future developments may involve deep learning for aerospace alloy design, real-time material monitoring, and AI-assisted smart manufacturing systems. Materials Informatics is expected to play a major role in next-generation aerospace engineering.

14. CONCLUSION

This project successfully demonstrated the application of machine learning in predicting and optimizing aerospace aluminum alloy strength. Linear Regression effectively predicted tensile strength using alloy composition and processing parameters.

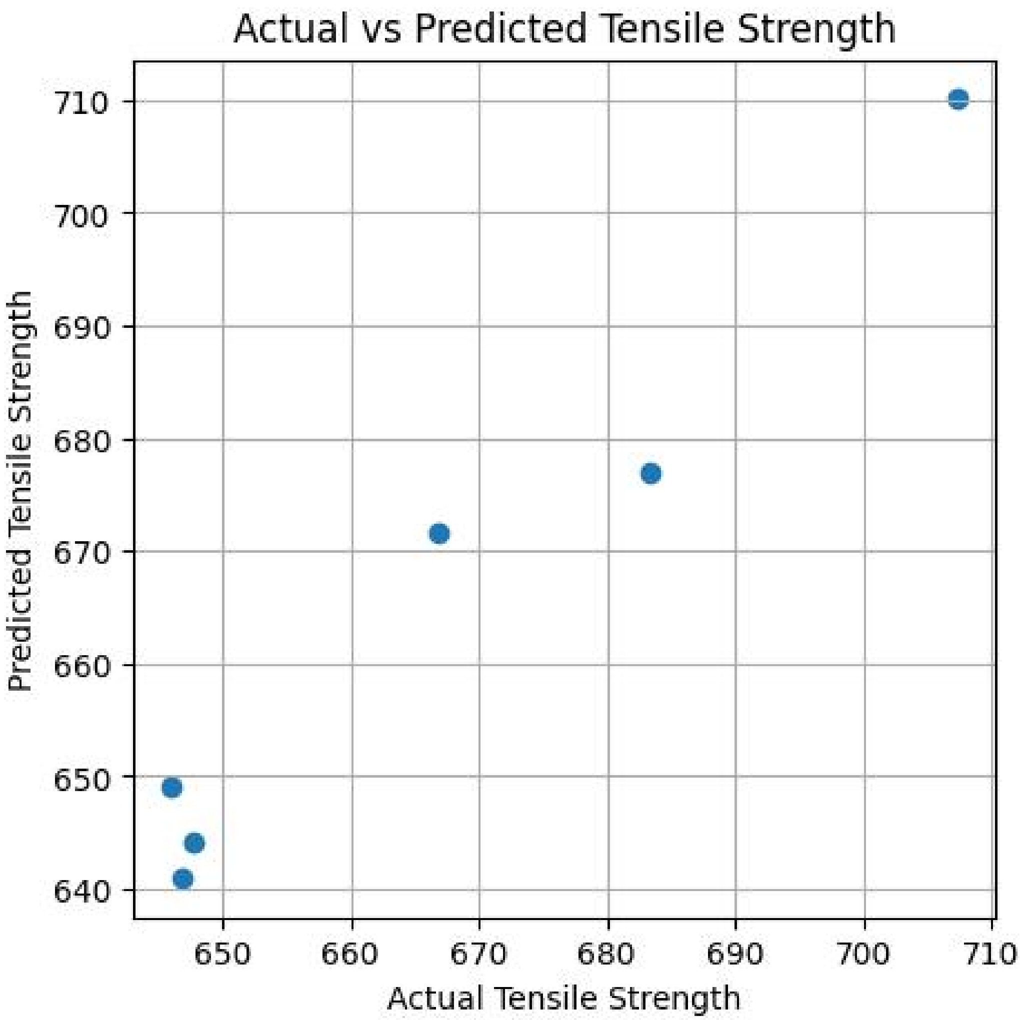
The results showed that machine learning can significantly accelerate aerospace material development while reducing cost and experimental effort. The optimized alloy composition proposed in this study provides improved strength and lightweight performance suitable for aerospace applications.

15. REFERENCES

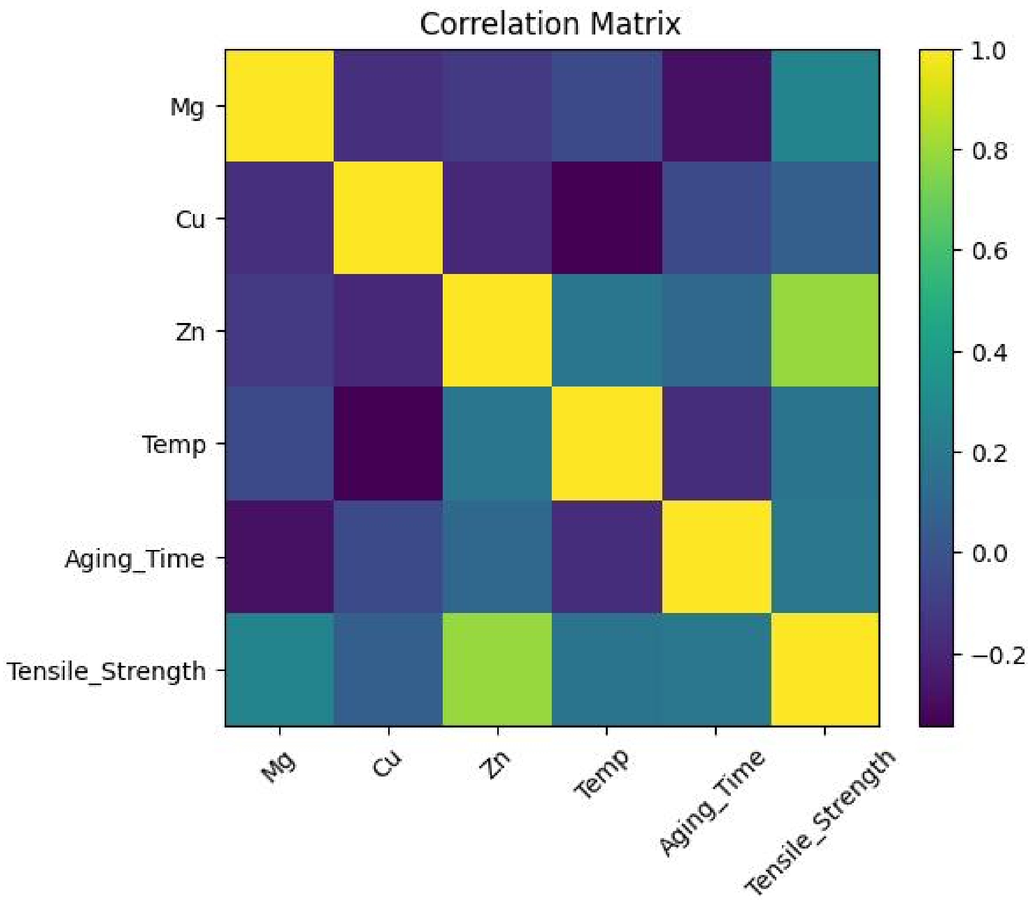
1. Callister, W.D., Materials Science and Engineering.
2. Smith, W.F., Foundations of Materials Science and Engineering.
3. Scikit-learn Documentation.
4. Aerospace Materials Handbook.
5. Python Machine Learning by Sebastian Raschka.
6. Research papers on aerospace aluminum alloys.
7. Materials Informatics assignment guidelines.

GRAPHS AND VISUALIZATIONS

Actual vs Predicted Tensile Strength Plot



Correlation Matrix



SAMPLE DATASET

Mg	Cu	Zn	Temp	Aging_Time	Tensile_Strength
2.37	1.73	5.58	122.0	11.0	650.3
2.95	1.42	5.41	132.0	9.0	658.65
2.73	1.35	6.24	131.0	9.0	689.69
2.6	1.96	5.54	120.0	12.0	667.65
2.16	1.98	5.42	118.0	13.0	662.73
2.16	1.87	5.81	122.0	8.0	654.13
2.06	1.51	5.21	140.0	12.0	633.17
2.87	1.37	6.2	131.0	13.0	701.46
2.6	1.78	5.11	124.0	11.0	646.03
2.71	1.61	6.48	126.0	11.0	707.3